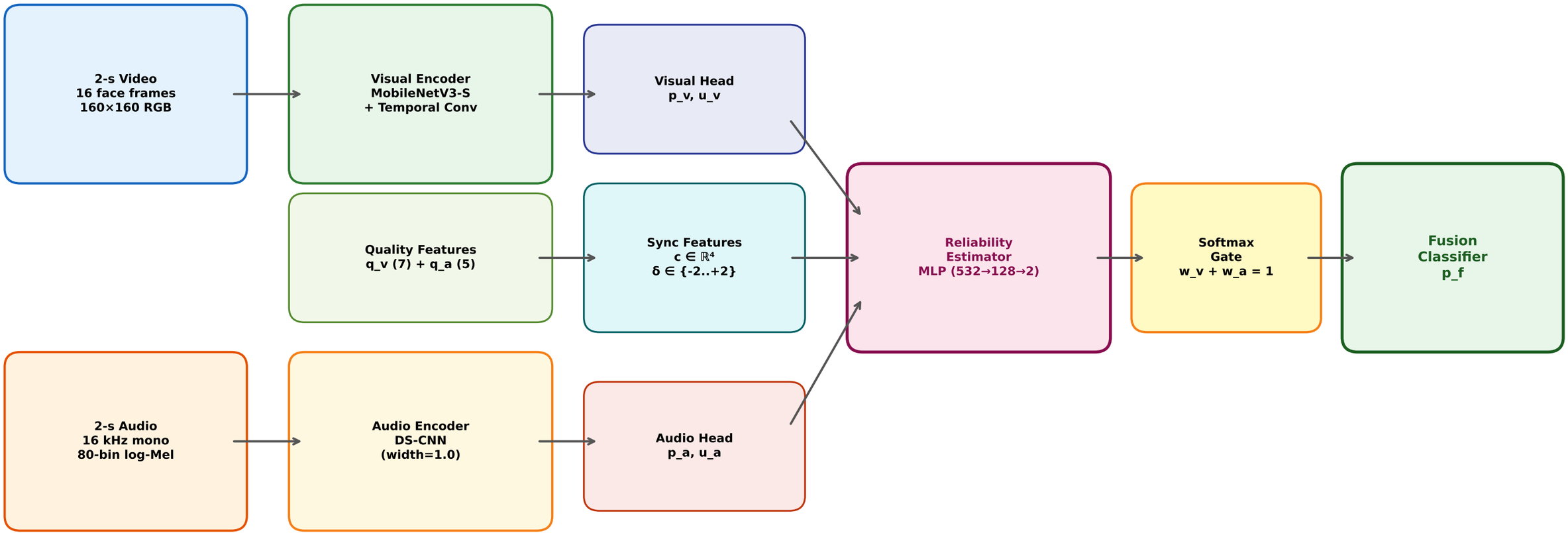
System Architecture: Reliability-Aware Audio-Visual Deepfake Detector
2-s Video
16 face frames
160×160 RGB
Visual Encoder
MobileNetV3-S
+ Temporal Conv
Visual Head
p_v, u_v
Quality Features
q_v (7) + q_a (5)
Sync Features
c ∈ ℝ⁴
δ ∈ {-2..+2}
2-s Audio
16 kHz mono
80-bin log-Mel
Audio Encoder
DS-CNN
(width=1.0)
Audio Head
p_a, u_a
Reliability
Estimator
MLP (532→128→2)
Softmax
Gate
w_v + w_a = 1
Fusion
Classifier
p_f